

POWER BI DASHBOARD DOCUMENTATION

Olist BI Retail Project

Business Intelligence & Data Analytics

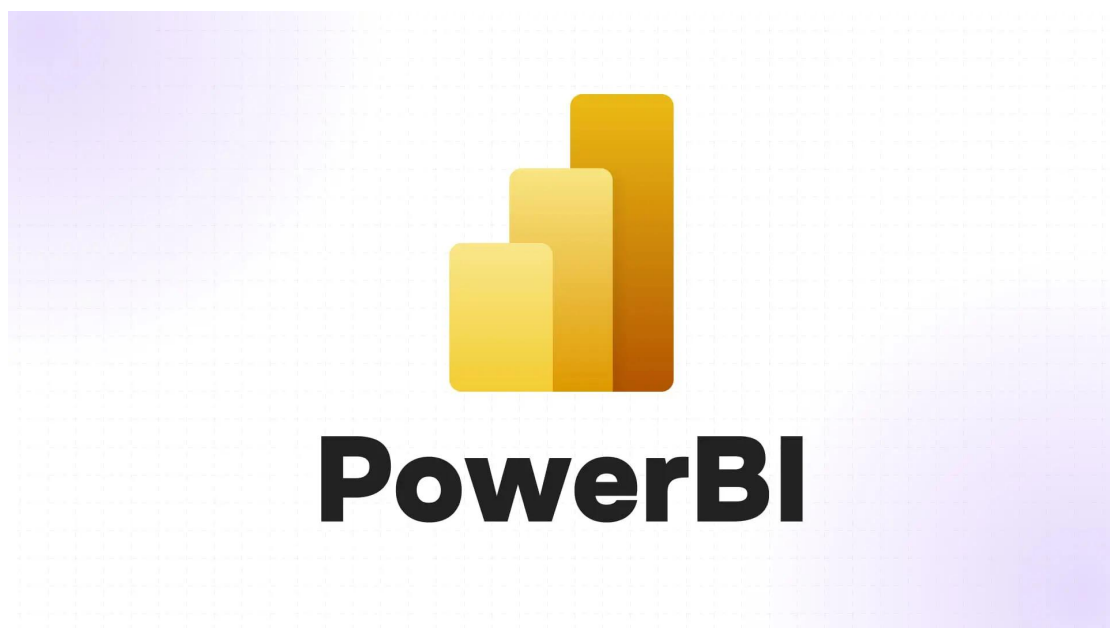
From Data Warehouse to Interactive
Business Insights

Technology: Microsoft Power BI

Data Source: PostgreSQL Data Warehouse

Data Model: Star Schema

Project: Olist E-Commerce Analytics



1. Introduction

Microsoft Power BI was used as the Business Intelligence and data visualization layer of the Olist BI Retail project.

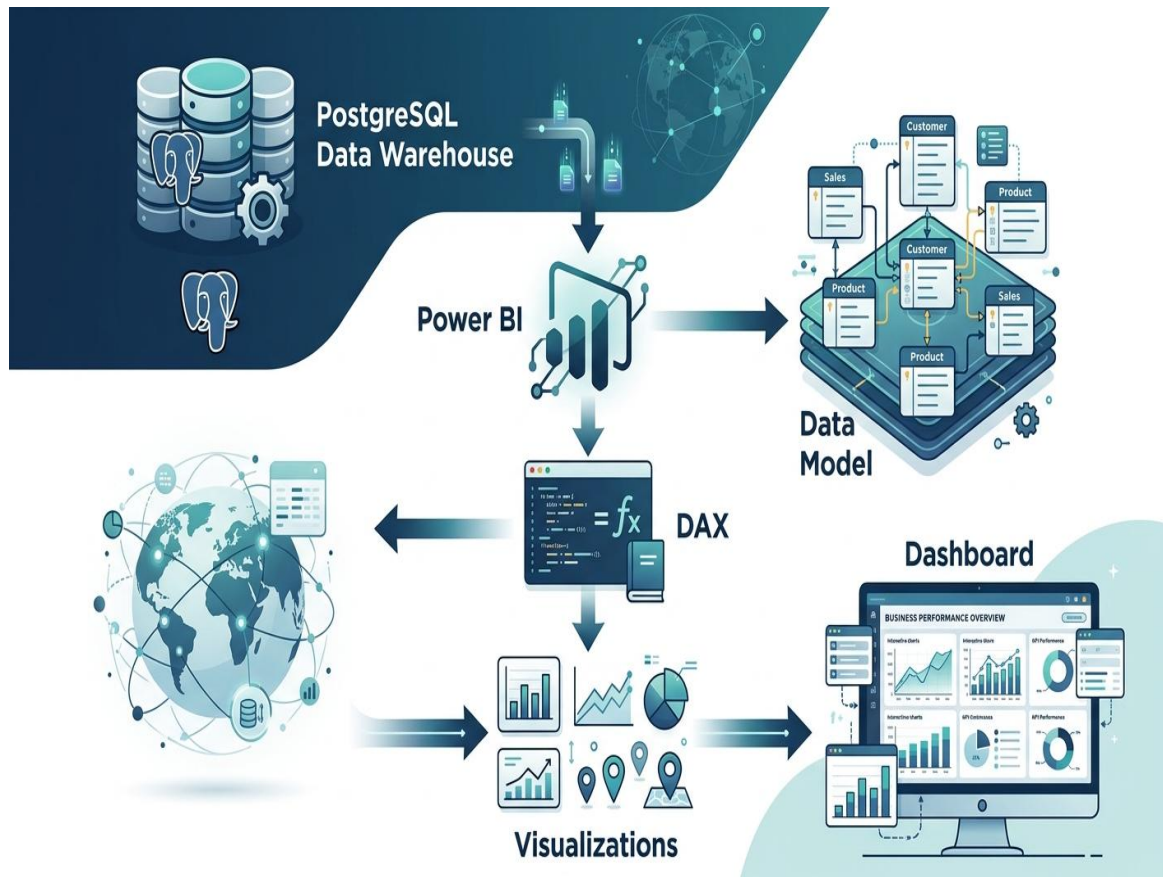
The objective of this stage is to transform the structured data stored in the PostgreSQL Data Warehouse into an interactive analytical environment that enables users to monitor business performance, identify trends, compare different business entities, and support data-driven decision-making.

The Power BI solution integrates the Data Warehouse, data model, DAX measures, interactive visualizations, filters, and navigation into a unified analytical dashboard.

Main objectives

- Monitor overall business performance.
- Analyze sales trends and revenue.
- Understand customer distribution and behavior.
- Evaluate seller performance.
- Analyze logistics and delivery performance.

2. Power BI Architecture



The architecture separates data engineering activities from business intelligence activities.

Data Engineering Layer

The raw Olist datasets are extracted, transformed, validated, and loaded using Talend into PostgreSQL.

Data Warehouse Layer

The transformed data is organized using a **Star Schema**, providing a structured and optimized source for analytical queries.

Business Intelligence Layer

Power BI connects to the Data Warehouse and provides:

- Data modeling
- DAX calculations
- Interactive visualizations
- KPI monitoring
- Filtering
- Navigation
- Business analysis

3. Data Source and Power BI Connection

Power BI is connected to the PostgreSQL Data Warehouse developed during the previous stages of the project.

Source

Database: PostgreSQL

Layer: Data Warehouse

Schema: Data Warehouse

The Power BI model uses the curated tables rather than directly consuming the original CSV files.

This approach provides a clear separation between:

Raw Data → ETL → Data Warehouse → BI

Main tables used

- fact_sales
- dim_customer
- dim_product
- dim_seller
- dim_order

- dim_date
- dim_payment

Objective

Using the Data Warehouse as the Power BI source ensures that:

- data is already transformed;
- business entities are clearly structured;
- relationships are controlled;
- analytical calculations are easier to maintain;
- the dashboard is based on a centralized source of truth.

4. Data Model

The Power BI data model follows a **Star Schema** architecture.

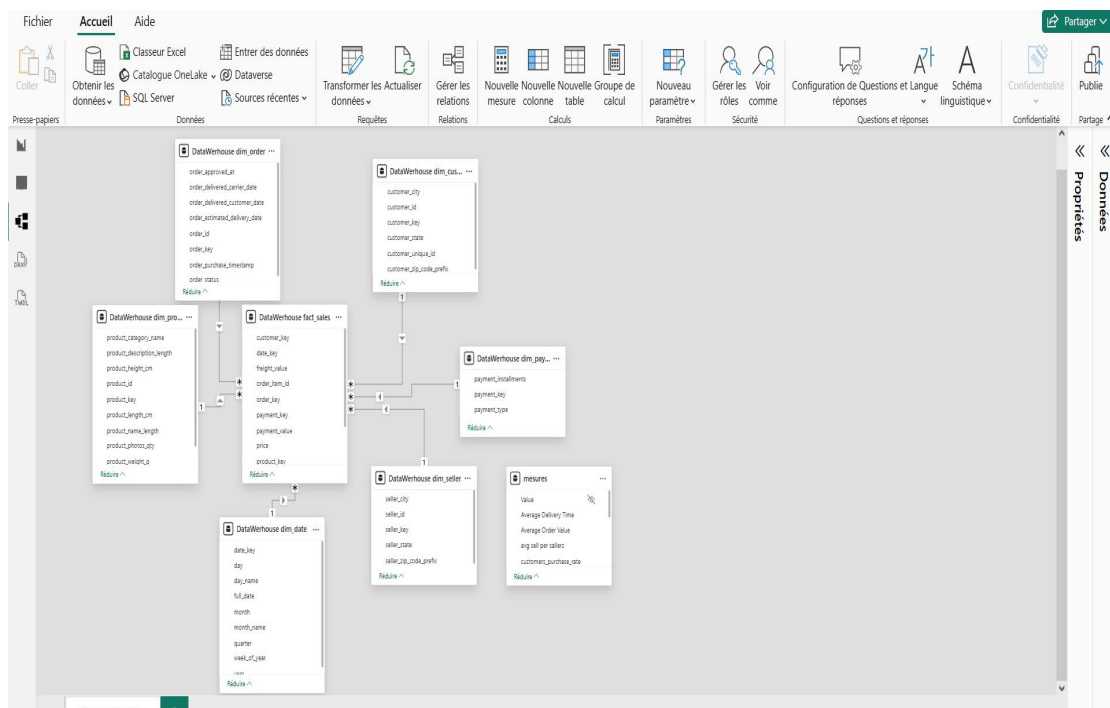
The central table is:

fact_sales

It contains transactional information related to sales, including:

- Order
- Product
- Seller

- Customer
- Price
- Freight Value
- Payment Value
- Sales-related measures



Dimension tables

dim_customer

Contains customer information and geographic attributes.

dim_product

Contains product information and product categories.

dim_seller

Contains seller information and geographic attributes.

dim_order

Contains order-related information and order status.

dim_date

Provides the temporal dimension required for time-based analysis.

dim_payment

Contains payment-related information

5. Data Modeling

The relationships between the fact and dimension tables allow Power BI to filter and aggregate sales data according to different business perspectives.

The model follows the principle:

Dimensions provide the analytical context, while the fact table provides the measurable business events.

Dimension	Analytical Perspective
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Customer	Customer analysis
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Product	Product/category analysis
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Seller	Seller performance
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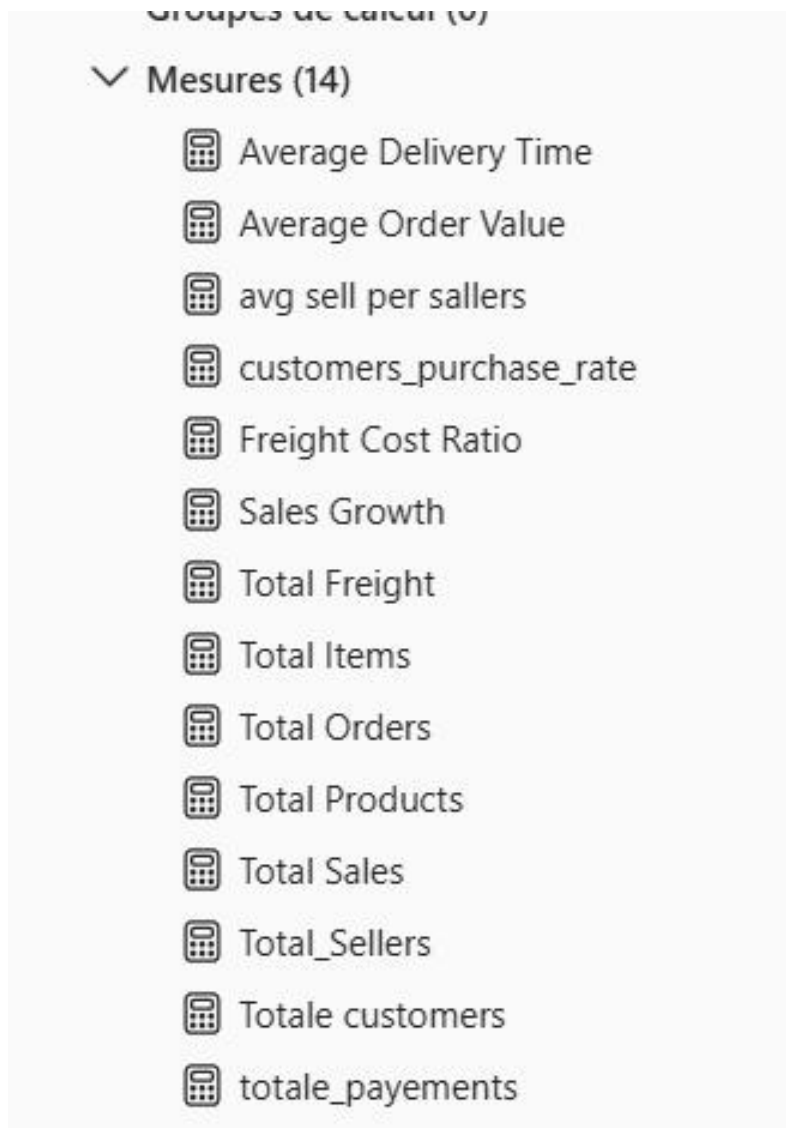
Order	Order status and order analysis
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Date	Time-based analysis
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Payment	Payment analysis
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6. DAX Measures

DAX (Data Analysis Expressions) was used to create the main business metrics required for the dashboard.



8. Dashboard Design

The dashboard was designed according to a **clean, consistent, and business-oriented visual approach**.

Design principles

- Consistent color palette.
- Clear KPI cards.
- Limited visual clutter.
- Consistent typography.
- Structured page layout.
- Simple navigation.
- Clear chart titles.
- Consistent spacing and alignment.

9. Executive Overview

The **Executive Overview** provides a high-level view of the overall business performance.

Main KPIs

- Total Sales
- Total Orders
- Total Customers
- Average Order Value

Main visualizations

Sales Performance Over Time

Shows the evolution of sales over the analyzed period.

Orders by Status

Provides an overview of the distribution of orders according to their status.

Sales by Product Category

Shows the contribution of different product categories to total sales.

Business objective

The Executive Overview is designed to provide decision-makers with a quick understanding of the overall performance of the e-commerce business.

 Insert screenshot of your Overview page here.

Page 11 — 10. Sales Analysis

10. Sales Analysis

The Sales Analysis page focuses on understanding revenue generation and sales performance.

Main KPIs

- Total Sales
- Total Orders
- Average Order Value
- Total Freight

Main analyses

- Monthly Sales Trend
- Sales by Product Category
- Sales by State
- Sales performance over time

Business objective

This page helps identify:

- sales trends;
- periods of high and low performance;
- important product categories;

- geographic sales distribution;
- changes in revenue performance.

 Insert screenshot of Sales page here.

Page 12 — 11. Customer Analysis

11. Customer Analysis

The Customer Analysis page focuses on understanding the geographic distribution and contribution of customers.

Main KPIs

- Total Customers
- Total Orders
- Average Order Value
- Total Sales

Main visualizations

Top 10 Cities by Customers

Identifies the cities with the highest number of customers.

Customers by State

Shows the geographic distribution of customers across Brazilian states.

Customer Geographic Distribution

Provides an overview of where customers are concentrated.

Business objective

The analysis helps identify major customer markets and geographic areas with high customer concentration.

 Insert screenshot of Customer Analysis here.

Page 13 — 12. Seller Analysis

12. Seller Analysis

The Seller Analysis page evaluates seller performance and activity.

Main KPIs

- Total Sellers
- Total Sales
- Total Orders
- Average Sales per Seller

Main visualizations

Top 10 Sellers by Sales

Identifies the sellers generating the highest sales.

Top 10 Sellers by Orders

Identifies the sellers handling the highest number of orders.

Business objective

This page helps compare sellers according to both:

- **financial performance**
- **order volume**

This provides a more complete view of seller performance.

 Insert screenshot of Seller Analysis here.

Page 14 — 13. Logistics Analysis

13. Logistics Analysis

The Logistics Analysis page focuses on delivery performance and transportation costs.

Main KPIs

- Total Orders
- Total Freight
- Average Freight per Order
- Average Delivery Time

Main visualizations

Average Delivery Time

Measures the average number of days required to deliver an order.

Freight Value by State

Shows the distribution of freight costs across geographic areas.

Business objective

The logistics dashboard helps identify:

- delivery performance;
- transportation costs;
- geographic areas with high freight costs;
- potential logistics optimization opportunities.

 Insert screenshot of Logistics Analysis here.

Page 15 — 14. Interactivity & User Experience

14. Interactivity and User Experience

Power BI provides an interactive analytical experience rather than a static report.

The dashboard allows users to:

- navigate between pages;
- filter data;
- interact with visualizations;
- explore different business dimensions;
- analyze data across different time periods.

Interactive navigation

The navigation buttons provide quick access to:

Overview → Sales → Customers → Sellers → Logistics

Filtering

Date and geographic filters can be used to focus the analysis on specific periods or areas.

Cross-filtering

Selecting an element in one visualization can dynamically affect other visuals when the relationships and interactions are configured correctly.

Page 16 — 15. Business Insights

15. Business Insights

The Power BI dashboard transforms the Data Warehouse into a decision-support environment.

The dashboard enables users to identify:

Sales Insights

- Revenue trends over time.
- Best-performing product categories.
- Geographic sales distribution.

Customer Insights

- Main customer markets.
- Customer concentration by city and state.
- Geographic opportunities.

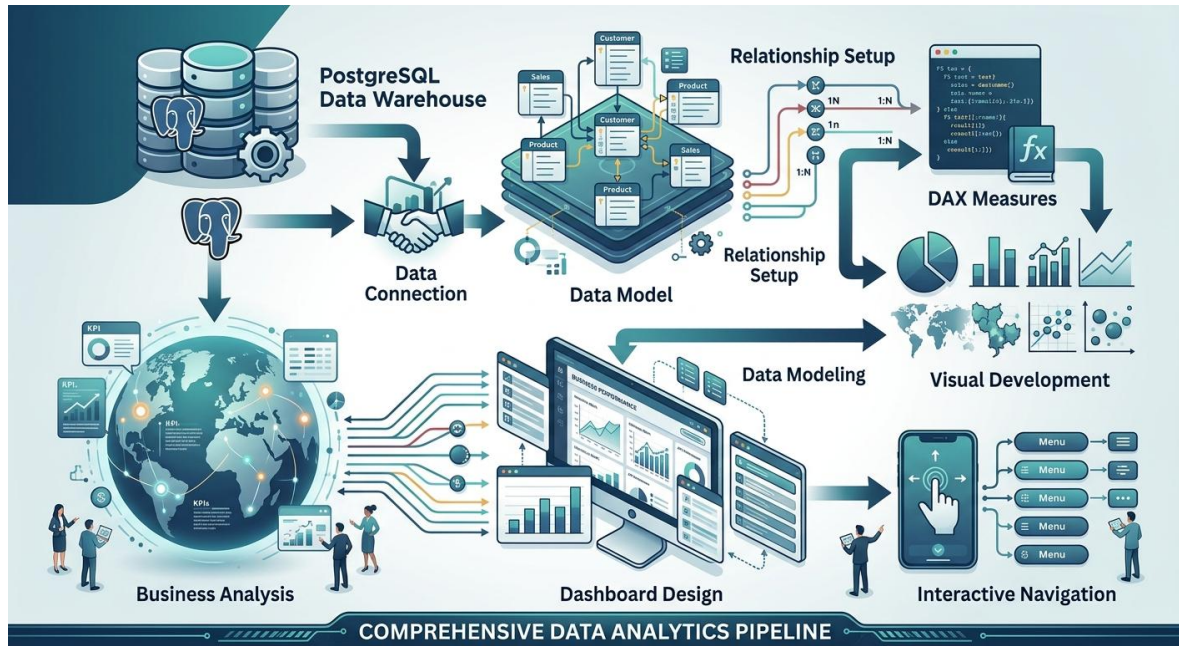
Seller Insights

- Top-performing sellers.
- Sellers generating the highest order volumes.
- Differences between sales performance and order activity.

Logistics Insights

- Average delivery performance.
- Freight cost distribution.
- Geographic areas with higher transportation costs.

These insights can support strategic decisions related to sales, customer targeting, seller management, and logistics optimization.



Conclusion

The Power BI implementation successfully transformed the Olist Data Warehouse into an interactive Business Intelligence solution.

The developed dashboard provides five complementary analytical perspectives covering **overall business performance, sales, customers, sellers, and logistics**.

The combination of a structured Star Schema, reusable DAX measures, interactive visualizations, and consistent dashboard design provides a reliable environment for exploring business performance and supporting data-driven decision-making.

Power BI therefore represents the final analytical layer of the BI pipeline, connecting the technical work performed during the **ETL and Data Warehouse stages** with actionable business insights.